SAMPLE

# Great Observatories: Swift Observatory

Catching Gamma-Ray Bursts

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Free sample: Introduction & Chapter 1

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Introduction

A small telescope with three eyes rides a low orbit around the Earth, and every time the sky flashes in gamma rays, it turns to look before the flash has faded. This is the Swift Observatory, launched by NASA in November 2004, renamed the Neil Gehrels Swift Observatory in 2018 for the scientist who led it from its first sketches, and still, at this volume's writing, the world's standing sentry for the most violent events the universe can stage: the gamma-ray bursts, the fleeting explosions that outshine, for a few seconds, everything else in the observable sky combined. No single instrument has done more to turn the bursts from a mystery catalogued at the edge of detection into a working branch of astronomy, and the observatory's story is one of the modern era's best tales of a modest machine answering enormous questions.

The design is the story. A traditional telescope that sees a burst can only wish it had looked sooner, because the bursts fade in seconds and the earlier orbiting instruments could not turn; the Swift team built a spacecraft that could, a nimble observatory whose Burst Alert Telescope watches a sixth of the sky at a time from behind a coded mask, whose onboard brain decides within seconds that a new source has appeared, and whose two narrow-field telescopes, an X-ray and an ultraviolet optical instrument, ride a rotating frame that swings the whole craft onto the new target in about a minute. The speed did what the previous thirty years of burst astronomy could not: it caught the afterglow, the fading X-ray, optical, and radio glow that follows the initial flash, while the glow was still bright enough to measure, to locate to a galaxy, and to name.

What the speed found fills the chapters ahead. The long bursts, the team confirmed, are the deaths of the most massive stars, each burst a supernova's inner engine; the short bursts, which the older instruments could barely even locate, proved to be the mergers of compact dead stars, neutron stars spiraling into neutron stars, a finding sealed in 2017 when a gravitational wave observatory's chirp, a gamma-ray flash, and a Swift-class follow-up all came from the same event; and along the way the sentry picked up unlooked-for prizes, the first great tidal disruption caught in X-rays as a black hole swallowed a star, the flares of magnetars, and a decade's archive of the quiet X-ray sky.

The volume tells the machine's biography in the series' chapters, from the drawing boards of the 1990s through the launch and the first decade of science to the observatory's standing place in the modern network, and the reader who closes it will understand how a telescope the size of a small car, in an orbit lower than the space station's, opened the hardest window the sky

has.

A word about the record stands at the head of the account, since the reader should know what the machine's biography rests on. The mission's own documents are the primary source: the design reports and the instrument papers of the proposal and build era, the launch and commissioning records of 2004 and 2005, the mission's operations logs and the public health reports of the running years, and the burst catalogs the team has published across the two decades; the science's literature is the second layer, the thousands of papers the community has built on the observatory's alerts and data; and the modern histories of the burst field, written by the participants and the historians, supply the context and the verdicts. The record of a working spacecraft is unusually rich, richer than any earlier telescope's, since the modern missions document themselves in real time, and the biography's every chapter stands on that open record, which the reader can consult entire.

The burst field's middle years, between the Vela discovery and the afterglow era, deserve their own paragraph, since the observatory's design was their direct answer. The theories of the 1980s, the galactic neutron star camp and the cosmological camp, fought over samples of dozens of bursts, the largest any instrument had gathered; the distance question's resolution waited on positions the gamma-ray detectors could not give; and the era's best instruments, the dedicated burst monitors of the late 1980s and the great observatories pressed into the watch, each contributed a piece, the sky map's rough isotropy, the population's rough bimodality, and the source counts' cosmological hint, without breaking the deadlock; the era's lesson, which the observatory's generation took as their founding brief, is the standing one of the astronomy's instrumentation: a science limited by its instrument's positional accuracy should build a better instrument, and the burst field's better instrument was the fast telescope with the coded mask, proposed as the 1980s closed and flying as the 2000s opened.

One further piece of the origin story belongs in the chapter's close, the mission's own stated purpose, quoted from the proposal's language the field's histories preserve. The burst problem, the founding document said, required a telescope that could see the whole of a burst's observable life, from the flash to the fading, in the first minutes no instrument had ever seen; the observatory was proposed as exactly that, the first telescope of the burst's whole life, and the purpose's plainness is the origin story's own close: the machine was designed, from the first sketch to the launch pad, to arrive in time, and the arrival in time, as the chapters have told, is the whole of the discovery the

follows.

Chapter 1: The Discovery of Swift Observatory

The origin story of a machine is this chapter's discovery, and it begins with a problem nobody could solve.

The problem was the bursts themselves, and their discovery's date belongs to an earlier era: 1967, when the American Vela satellites, built to watch the Earth for the secret tests of nuclear weapons, recorded instead a series of flashes in gamma rays from no direction on the planet. The flashes came from the sky, they lasted from fractions of a second to minutes, and they never repeated in place; the detection's announcement, held for years while the team ruled out the weapons they were built to catch, reached the journals in 1973, and the bursts entered astronomy as a pure mystery, energetic, brief, and unlocatable.

Unlocatable was the operative word, and it held for two decades. The gamma-ray detectors of the era measured the bursts' arrival with no precision to speak of, and the error boxes, the patches of sky the instruments could say the burst had come from, were larger than the full moon, sometimes larger than whole constellations; the optical and X-ray searches of the 1970s and 1980s, hunting a fading counterpart inside boxes that size, found nothing, and by the early 1990s the field had divided into a camp that expected the bursts to be local, some new flavor of event in the galaxy's own stars, and a camp that dared the enormous alternative, that the bursts were at cosmological distances and were, briefly, the brightest things the universe had ever shown.

The distance question broke in the 1990s, by a combination the chapters ahead honor properly. The Italian Dutch BeppoSAX satellite, carrying X-ray telescopes that could turn, caught the X-ray afterglow of a burst in 1997, the first afterglow ever seen; the follow-up from the ground, in that same month, found the fading optical glow and then the host galaxy, and the galaxy's redshift put the burst billions of light years away, in the remote universe, and the cosmological camp had its verdict. The bursts were the deaths of something enormous, at distances where the flash, to be seen at all, must outshine galaxies; and the afterglow, the discovery that unlocked the door, was found to fade in hours, which meant that the next steps of the science belonged to whichever telescope could arrive first.

The design for that telescope was already on paper, and its lineage is the chapter's second half. The NASA of the 1990s, hunting a middle-sized mission for the burst problem, received the proposal of a team at the Goddard Space Flight Center led by the gamma-ray astronomer Neil Gehrels, a veteran of the

field's earlier satellites; the proposal, selected in 1999 as a Mid-scale Explorer, described exactly the machine the problem demanded, a wide-field burst monitor for the alert, a fast-slewing spacecraft for the response, and two follow-on telescopes for the afterglow, with the decisions made onboard, by the spacecraft itself, because the light lag to the ground controllers would cost the minutes the afterglow could not spare.

The mission took the name of the swift, the bird, chosen for the machine's single defining trait, and the name's aptness the team underlined with a logo of the bird in flight; the observatory was built across the familiar American aerospace landscape, the instrument teams at Goddard, at Penn State, at Leicester in England, and at the Brera observatory in Italy, the spacecraft assembled by Spectrum Astro in Arizona, and the whole tested against a deadline the science itself justified, since every year of delay was a year of afterglows lost.

The launch closed the origin story, cleanly, on 20 November 2004, when a Delta II rocket lifted the observatory from the Cape into a low circular orbit some six hundred kilometers up, a path chosen for the quick slewing and the low radiation, circling the planet every ninety odd minutes; the commissioning weeks went by the book, the instruments were tuned, the first test bursts were caught, and the observatory that had been a proposal for a decade opened its eyes as the fastest telescope the space age had flown. The chapters ahead tell what those eyes saw.

A note on the numbers the design runs on, offered once so the chapters can use them freely. The orbit: a low circle at roughly six hundred kilometers, an orbital period of about ninety minutes, a low inclination chosen for the launch site's reach and the quick data passes; the mass: a couple of thousand kilograms, the spacecraft and its instruments; the slewing: the full reorientation in about a minute, the fastest of any astrophysics mission flown; the watching: the Burst Alert Telescope's roughly a sixth of the sky, observed continuously; and the response: the burst alerts out to the world within seconds of the onboard detection, the X-ray images inside the first minutes, the optical follow-up inside the hour. The reader who keeps those six numbers holds the machine's whole character in shorthand, and the chapters ahead use them without further ceremony.

One further idea, the survey's stealth, belongs in the set, since the observatory has always been two telescopes in one. The Burst Alert Telescope's continuous watch, intended for the bursts, is also a hard X-ray survey of the

whole sky: the mask's view sweeps everything as the orbit precesses, and the

archive of the steady sky, accumulated across the two decades, is a deep map of the hard X-ray universe, the black holes' and the neutron stars' populations, the accreting binaries' long term light, and the quiet sky's standing census. The idea's lesson is the instrument designer's own: the monitor built for the transient is the survey of the steady, and the observatory's second archive, the map nobody planned, is the idea's standing product, which the modern surveys now mine.

The reader should also meet the design's political idea, since the observatory was an argument in a debate. The great observatories of the era, the flagship telescopes, were built for depth, the long looks and the faint sources, and the middle-class missions' argument was the other axis, the time: the universe's changes, the bursts' era had proven, held the biggest questions, and a machine that watched the changes would answer questions the deep looks could not ask. The observatory was the time-domain argument's flagship, the small fast telescope that out-discovered the decade's giants, and the design's idea, in the field's own telling, is the standing lesson of the modern astronomy's balance sheet, that the sky's stories are told in time as much as in depth, and the instrument that keeps both axes is the field's ideal, the successor generations' designs aimed at.

The final idea the chapter should hold is the design's humility, the engineering's own acknowledgment of the unknowns. The burst rate was uncertain when the observatory was proposed, and the team sized the instruments for the optimistic case while planning for the lean one; the afterglow's behavior was a theory's prediction, and the telescopes were built wide rather than narrow, to catch what the theory might miss; and the spacecraft's flexibility, the pointed program's share and the software's growth room, was the design's insurance against the science's surprises. The reader who follows the mission's history will see the humility vindicated: the surprises came, the tidal disruptions and the magnetars and the merger era, and the machine that had kept room for them caught them, the design's last idea being the field's first lesson, that the sky rewards the telescope that leaves space for what it did not expect.

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